K-FLATNESS OF A COMPLEX OF MODULES IS CHECKED ON STALKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the stalk computation; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-check-K-flat-stalks also occurs in sites-cohomology.tex. The ruling recorded for this lane before assembly is to keep this card: the sites statement is about a complex whose terms are flat modules over a ringed topos and its conclusion includes flatness of the terms, so the two are related by a genuinely stronger hypothesis and a genuinely stronger conclusion rather than being one statement restated.

Allowed standard prerequisites (editorial).

- Exactness of complexes of sheaves is checked on stalks. Modules, Lemma modules-lemma-abelian, invoked at proof lines 5885-5886. Supplies the first direction.
- Definition of K-flatness for complexes over a ring. More on Algebra, Definition more-algebra-definition-K-flat, named in the statement at line 5878. Supplies the property tested on stalks.
- Skyscraper pushforward along a point and its interaction with stalks and tensor products. Standard sheaf theory, used at proof lines 5889-5893. Supplies the acyclic complex of sheaves used to transfer the K-flat property from the space to the local ring.

Source reference. Original lemma 01AG: sources/stacks-chow/tags.txt, source lines 1689–1689. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 06XZ: sources/stacks-chow/tags.txt, source lines 8868–8868. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{K}^\bullet$ be a complex of $\mathcal{O}_X$-modules. Then $\mathcal{K}^\bullet$ is K-flat if and only if for all $x \in X$ the complex $\mathcal{K}^\bullet_x$ of $\mathcal{O}_{X,x}$-modules is K-flat (More on Algebra, Definition 06XZ).*

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Proof. If $\mathcal{K}_x^\bullet$ is K-flat for all $x \in X$ then we see that $\mathcal{K}^\bullet$ is K-flat because $\otimes$ and direct sums commute with taking stalks and because we can check exactness at stalks, see Modules, Lemma 01AG. Conversely, assume $\mathcal{K}^\bullet$ is K-flat. Pick $x \in X$ and let $M^\bullet$ be an acyclic complex of $\mathcal{O}_{X,x}$-modules. Then $i_{x,*}M^\bullet$ is an acyclic complex of $\mathcal{O}_X$-modules. Thus $\mathrm{Tot}(i_{x,*}M^\bullet \otimes_{\mathcal{O}_X} \mathcal{K}^\bullet)$ is acyclic. Taking stalks at x shows that $\mathrm{Tot}(M^\bullet \otimes_{\mathcal{O}_{X,x}} \mathcal{K}_x^\bullet)$ is acyclic. $\square$